

Effect of boron insertion on metal-film wetting of Fe on MgO

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Abstract

The wetting of a metal film on MgO(001)—the interface at the heart of MgO-based magnetic tunnel junctions—is governed by the competition between a flat, well-wetting film and a dewetted island. We map this competition for the same Fe host with and without added boron, using a biased, surrogate-driven search that targets the two configurations the wetting question is about. In both models the island is the ground state, so the flat film is a distinct, higher-energy basin—by 0.1888 eV/atom for Fe/MgO and by 0.1493 eV/atom for Fe-B/MgO. Adding boron lowers the flat-island separation by 0.040 eV/atom, a 21 % reduction, at a search-level significance of $p = 0.0444$, and boron acts inside the film rather than at the interface. Boron therefore moves the flat, well-wetting configuration closer to the island ground state without displacing it—the direction the confusion principle of metallic-glass formation predicts—and so favours the flat film that MgO-based tunnel junctions require.

1 Introduction

MgO-based magnetic tunnel junctions reach very large tunnel magnetoresistance at room temperature when the barrier is crystalline and the metal electrodes are matched to it in-plane [1, 2]. The transport calculations that account for the effect were carried out for an ideal Fe|MgO|Fe(001) interface, in which a flat metal layer sits directly above the substrate oxygen [3]. Achieving those values in practice is a question of realising that interface closely enough: the high magnetoresistance is obtained when the barrier is (001)-oriented and the metal film flat and continuous [1], and the barrier’s orientation and stress relaxation are themselves sensitive to how the metal is deposited and annealed [4, 5]. The structural quality of the metal/oxide interface is therefore part of the specification of the device rather than an incidental detail.

Two recent trends in junction fabrication have only sharpened that point. The first is the push on the tunnel magnetoresistance itself: after a decade of stagnation at the 604 % reported in 2008 for junctions with boron-alloy electrodes [5], epitaxial junctions now reach a record 631 % at room temperature (1143 % at 10 K), and the gain comes explicitly from the interface—from tuning the crystallographic orientation and the oxygen content of the MgO barrier through ultrathin metal insertions at the metal/oxide boundary [6]. The second is the move to perpendicularly magnetised junctions for dense, non-volatile memory, in which the metal/oxide interface is itself a functional layer: the Fe/MgO interface carries a large perpendicular surface anisotropy, whose strength is traced to the hybridisation of interfacial iron and oxygen states [7], and sub-nanometre metal films grown on the barrier are engineered for the perpendicular anisotropy and the low magnetic damping that low-power switching requires [8]. Both trends converge on the same requirement—a metal film that is flat, epitaxial and well-wetting on the crystalline barrier—and, in practice, on layer-by-layer control of the roughness and thickness of each film in the stack [9].

The flat, lattice-matched film assumed in such interface models is not, however, the configuration that a metal film necessarily adopts when it is deposited on MgO(001). At the monolayer coverages relevant to a thin electrode, growth experiments find the film dewetting into three-dimensional islands: He-atom scattering records three-dimensional island growth of Fe on MgO(001) at room temperature, suppressed only when the film is deposited at 140 K [10]; scanning tunnelling microscopy combined with X-ray magnetic circular dichroism finds sub-nanometre Fe growing three-dimensionally, with a two-dimensional growth mode reported only above about 6.5 monolayers [11]; and grazing-incidence small-angle X-ray scattering on five

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Table 1: The two interface models: film constitution and total atom count.

Model	Composition	Film species	N atoms
Fe/MgO	$\text{Fe}_{25}\text{Mg}_{25}\text{O}_{25}$	Fe	75
Fe-B/MgO	$\text{B}_3\text{Fe}_{25}\text{Mg}_{25}\text{O}_{25}$	Fe + B	78

monolayers likewise identifies Volmer–Weber growth with spherical islands [12]. A flat, pseudomorphic monolayer is obtained instead by slow deposition onto a cleaved, oxygen-annealed crystal [13]. The islanding tendency survives even into the most controlled modern fabrication: grain-to-grain epitaxial growth of a sub-nanometre ferromagnetic layer on a polycrystalline MgO(001) barrier on 300 mm wafers must be performed at cryogenic temperature (100 K), because room-temperature deposition lets the metal nucleate as islands and break the film’s continuity [8]. Two structural outcomes thus compete at this coverage—a flat, well-wetting film and a dewetted island—and the growth experiments constrain their kinetics rather than their relative energies, while interface models assume the flat geometry by construction. What is missing is a comparison of the two configurations’ energies for one and the same interface, and an answer to how that comparison responds to a change in the film’s composition.

The composition is not arbitrary. The metal electrodes of the highest-magnetoresistance MgO junctions are boron-bearing and amorphous as deposited, which is also the condition under which the MgO barrier grows (001)-textured [4]. Boron is a glass-forming addition, and added species are in general understood to stabilise disordered configurations at the expense of crystalline ones, in the core of the confusion principle of metallic-glass formation [14]. Applied to this interface, where the flat film is the two-dimensional, disordered-like configuration and the island the three-dimensional, ordered-like one, it suggests that boron should lower the energy of the flat configuration relative to the island. Whether it does so, and by how much, is a question for calculation.

Testing this idea directly is not a single-calculation question. The flat configuration is one well-defined structure, but the ordered island it competes with is not: the dewetted film comprises a large family of distinct structures with many local minima, so no one geometry can represent it, and the two phases cannot be compared by relaxing a single candidate of each. The confusion-principle question—whether an added element lowers the disordered configuration relative to the ordered one—is therefore about the landscape as a whole, and one that is expensive to answer, because exploring it with density-functional theory alone would require evaluating far too many structures. Two features of the problem make it tractable. First, the question is specifically about two phases—the flat, well-wetting film and the dewetted island—so rather than surveying every configuration we restrict the exploration to these two targets, a deliberately biased strategy. Second, because the search must still sample many candidates to populate both phases, we guide it with a surrogate model, so that the expensive density-functional evaluations are spent only where they are most informative.

In this work, by implementing a surrogate-driven active-learning search with a deliberately biased exploration strategy [15, 16], we categorize the potential-energy surface of Fe on MgO(001) into its two growth modes—the flat, well-wetting film and the dewetted island—for the same host with and without added boron, and quantify how the added element shifts the relative energy of the flat mode.

This paper is organized as follows. In Sec. 2, we present the active-learning search, the two interface models, and the metrics used to separate the two growth modes. In Sec. 3, we first establish the two-phase landscape and the island ground state, and then report the effect of boron on the flat–island separation. In Sec. 4, we discuss the origin of the island ground state and the role of the added element. We summarize our work in Sec. 5.

2 Methods

2.1 Interface models

We study the metal-on-oxide interface relevant to magnetic tunnel junctions: a thin metal film deposited on the MgO(001) surface. Two compositions were considered, differing only in whether boron is present in the film (Table 1)—the pair that isolates the effect of the added metalloid in a single host.

Table 2: Candidate-generation schedule, common to both models: a small-scale heterostructure-aware randomiser and a large-scale rattle generator. The number of candidates generated per phase is N .

Phase	Iterations	Small-scale	Large-scale	Total N
I	$0 \leq i < 10$	20	0	20
II	$10 \leq i < 25$	10	10	20
III	$25 \leq i$	0	20	20

The substrate is a single MgO(001) layer and the film a single metal layer, stacked along z with 20 Å of vacuum and periodic boundaries in-plane. The in-plane lattice of the simulation cell is the DFT-optimised Fe lattice constant ($a_{\text{Fe}} = 2.870$ Å), used in a $(5 \times 5 \times 1)$ supercell throughout. The MgO layer is built epitaxially onto that geometry: the substrate is assembled on the Fe(001) surface cell with the oxygen sublattice placed directly above the metal sites (a $\sqrt{2}$ rotation relative to the bulk MgO cell), so the MgO in-plane parameter is forced to 2.870 Å. Compared with the bulk value ($a_{\text{MgO}}/\sqrt{2} = 2.978$ Å) this means the MgO is compressed in-plane by 3.6 %, i.e. the mismatch is accommodated by the substrate, not by the film—the Fe film remains at its own equilibrium lattice constant and is not strained in-plane. The substrate is then held fixed in that compressed state and only the film is allowed to relax.

2.2 Biased global optimisation (AGOX / GOFEE)

Structures were sampled with AGOX [17], using a GOFEE-type surrogate-driven search [15]: a Gaussian-process regression (GPR) surrogate with an Oganov-type fingerprint descriptor [18] and a repulsive prior, coupled to a lower-confidence-bound (LCB) acquisition function ($\kappa = 2$). Each search ran 100 iterations.

Three-phase biased exploration

Candidate generation is not uniform over the run but proceeds in three phases selected by the iteration counter i , mixing generators of different perturbation scale. Both models combine a small-scale heterostructure-aware randomiser (rattle amplitude 1.5, all film atoms) with a large-scale rattle generator (half the film atoms, rattle amplitude 2.3); the schedule is identical for the two models and is given in Table 2.

The per-phase candidate total is 20; the two models differ only in whether the film contains boron, not in how candidates are generated. The resulting workflow is summarised in Fig. 1.

The early phases favour the small-scale (structure-aware) generator, which preserves the layer-resolved character of the flat reference basin, while the later phases shift to the large-scale rattle generator for broader exploration.

Bias

The exploration is deliberately biased: the search is seeded from a flat metal layer—the reference film geometry, a single-monolayer-thick film rather than a random three-dimensional distribution of metal atoms—so the sampled database is a mixture of the flat basin and any lower-lying basin the search finds. The composition of the reference layer follows the target stoichiometry (Sec. 2.1): a pure Fe layer for Fe/MgO, and the same Fe layer with B added above the surface for Fe-B/MgO. The films are single-layer in both models, and the two reference layers differ only by the boron. Each model was run from multiple independent random seeds, and each seed constitutes one independent search.

Only completed searches are used

Every number in this paper comes from searches that ran the full 100-iteration budget: 13 completed searches for Fe/MgO and 6 for Fe-B/MgO, 19 in total. Two further searches were started and are *not* used, and we record them here rather than drop them silently—one Fe/MgO search reached only iteration 37, and one Fe-B/MgO search produced no stored structures at all. A stopped search has had less search time than a completed one, so its best energy is systematically worse; and because such searches are not distributed

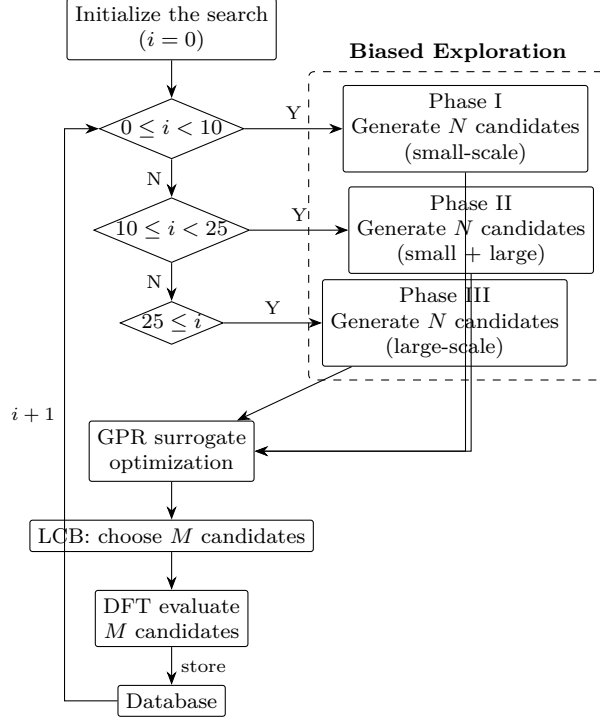


Figure 1: The biased-exploration loop. Each iteration selects a phase from the counter i , generates N candidates with that phase’s generator mix (Table 2), optimises them against the GPR surrogate, ranks them with the LCB acquisition, evaluates the selected candidates with DFT and stores them in the database; the counter then advances and the phase is re-selected.

evenly across the two models, including them would bias the comparison between them—as the method-parameter study in the Supplementary Material also shows, where searches that stopped early inflate the mean of one setting and depress another (Sec. S3 of the Supplementary Material). The same criterion is applied without exception throughout the analysis, and it is decided from the iteration count recorded for each search, not from how a search is labelled: a search that looks complete can have stopped early, and one whose stored structures are missing is treated the same way. In the two models used here the excluded searches happen to be identifiable at a glance; the rule does not rely on that.

Candidates were relaxed by the surrogate model for up to 100 steps (starting from iteration 10) and evaluated with the DFT calculator below. Only structures from iteration ≥ 10 were used in the analysis, i.e. after relaxation had begun; earlier pre-relaxation placements were discarded.

2.3 DFT settings

Energies and forces were obtained with GPAW [19] in LCAO mode (double-zeta polarised basis), the PBE exchange–correlation functional [20], a $(1 \times 1 \times 1)$ k-point mesh, Fermi–Dirac smearing (width 0.05 eV), and spin polarisation with Hund’s rule coupling enabled; a Pulay mixer and a fixed convergence criterion (energy 10^{-4} eV, density/eigenstates 10^{-3}) were used throughout. Note the deliberately modest computational setup (single-layer slab, Γ -point sampling): the study targets trends across compositions, not converged absolute energies (see Sec. 2.6).

The interfacial Fe–O separation is a construction parameter rather than a computed quantity: the reference films are built with the first oxygen layer of the substrate 2.3 Å above the metal plane, and relaxation preserves that spacing (2.300 Å for the flat basin, i.e. unchanged to the printed precision, and 2.331 Å for the island). Comparison with the 2.0–2.3 Å range reported for this interface by LEED analyses and earlier calculations [3, 13] therefore checks the model construction rather than testing the electronic-structure method. Independent of this, the LCAO basis set is appreciably less complete than a plane-wave or real-space rep-

resentation [21], so absolute geometric parameters are reported at fixed settings and are not interpreted as converged values.

2.4 Flatness and energy metrics

Two quantities describe each sampled structure:

Film flatness ΔZ —the vertical span of the metal film, $\Delta Z = z(\text{metal})_{\text{max}} - z(\text{metal})_{\text{min}}$, taken over the film’s metal atoms (Fe; boron is a metalloid and is not part of the flatness metric). $\Delta Z \approx 0$ is a flat, wetting film; large ΔZ is a 3D island (dewetted). A threshold of $\Delta Z \leq 1.0 \text{ \AA}$ separates the “flat” and “island” basins.

Relative energy dE/N —the per-atom energy above the system’s global minimum, $dE/N = (E - E_{\text{min}})/N$, where E_{min} is the lowest energy found for that composition and N the atom count. By construction the global minimum is 0 eV/atom; per-atom normalisation makes the two models directly comparable despite their differing atom counts (75 and 78 atoms).

For each model, the flat-basin ground state (lowest dE/N among $\Delta Z \leq 1.0 \text{ \AA}$ structures) was compared against the overall minimum, isolating the energy penalty of the flat configuration.

2.5 Robustness of the method

Because the search relies on a biased-exploration scheme, three method parameters were varied on the Fe/MgO model to test robustness (Supplementary Material): the rattle strength, the LCB parameter κ , and the inclusion of a dipole correction. The principal conclusion was shown to be insensitive to κ and to the dipole correction, and to degrade only when the rattle strength was reduced. This is the same model, method and reference layer as the Fe/MgO results of Sec. 3, so the study bounds the method used for the two models reported here.

2.6 Scope and limitations

The structures retained here are not DFT-converged minima: candidates were relaxed with the surrogate model and evaluated with a single DFT step, leaving residual forces of order 1–2 eV/Å. Accordingly, “lowest energy” and “basin” denote the lowest DFT energy *encountered* in the search, not a converged minimum, and the results are interpreted qualitatively (trends across compositions). The flat configuration is therefore described as a *higher-energy flat basin*, not as a metastable state—establishing metastability would require converged relaxations, a Hessian (no imaginary modes) and a barrier between basins. A full re-relaxation of representative structures is necessary for this purpose.

The models additionally build both phases on a body-centred-cubic Fe lattice, whereas the experimentally reported structure of ultrathin Fe on MgO(001) is body-centred tetragonal below about 10 Å, changing to bcc only above that thickness [13]. The island branch spans $\Delta Z \approx 1\text{--}6 \text{ \AA}$, so the structures compared here lie in the thickness regime in which the experimental film is not bcc. The flat–island comparison should therefore be read as a trend obtained within a fixed lattice model, not as a prediction of absolute structural parameters.

One further property of the model bounds its interpretation: the strain convention is the inverse of the experimental stack. The simulation cell takes the Fe lattice constant and the substrate is compressed to match it, so the Fe film is unstrained in-plane, whereas in a real junction a thin Fe film on bulk MgO absorbs the mismatch as in-plane strain and relieves it through interfacial dislocations. This model therefore does not represent the strained-film situation (see Sec. 4.4). The convention is the same for both models reported here, so it does not enter the Fe/MgO-vs-Fe-B/MgO comparison at all.

Finally, the two models differ *only* by the added boron: both are built on the same lattice constant, the same substrate, the same reference-layer geometry and the same candidate-generation schedule. What the comparison cannot separate is boron from *any* added element—a second, chemically different film composition would be needed for that, and none is analysed here.

Table 3: The two phases in each model ($\Delta Z \leq 1.0$ Å = flat/wetting; $\Delta Z > 1.0$ Å = island).

Model	Structures	Island	Flat	Global-min ΔZ (Å)	Flat-basin min dE/N (eV/atom)
Fe/MgO	1180	985	195	3.65	0.1888
Fe-B/MgO	543	430	113	3.77	0.1493

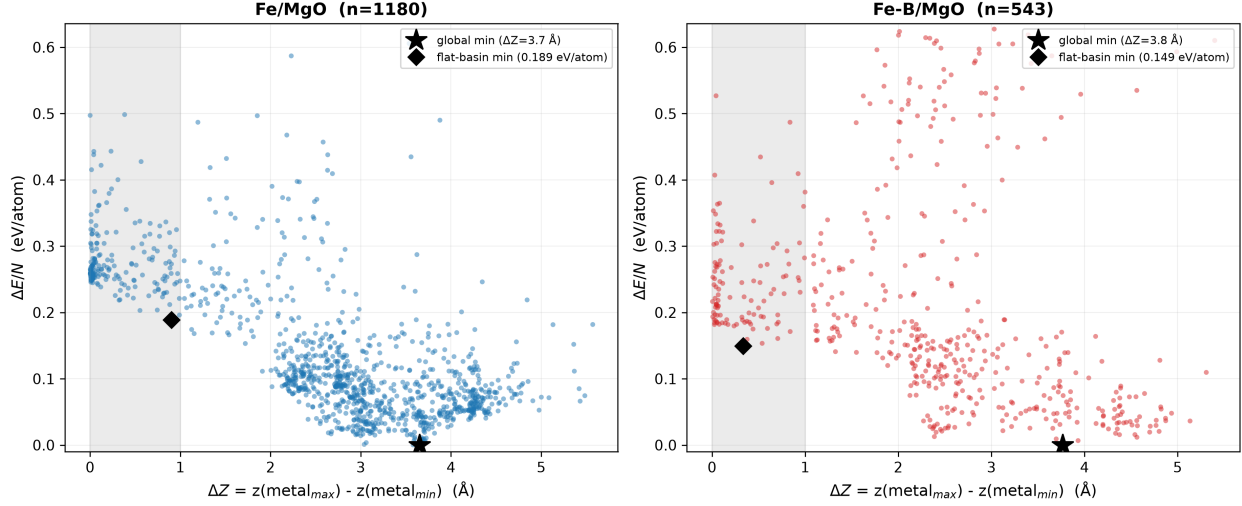


Figure 2: The sampled two-phase landscape of the metal film on MgO. Each point is one structure from the Fe/MgO (left) and Fe-B/MgO (right) searches, plotted as its relative energy dE/N above the model’s own lowest energy against the vertical span of the film, ΔZ . The shaded band is the flat window ($\Delta Z \leq 1.0$ Å); the star marks the model’s lowest-energy structure and the diamond the best member of its flat basin. Both panels use the same energy scale.

3 Results

3.1 The two-phase landscape of the metal film on MgO

Our target is the wetting of the metal film on MgO(001). The property that distinguishes a wetting from a dewetting film is its flatness, measured by the vertical span of the film, $\Delta Z = z(\text{metal})_{\text{max}} - z(\text{metal})_{\text{min}}$ over the film’s metal atoms (Fe); $\Delta Z \approx 0$ is a flat, wetting film and large ΔZ a three-dimensional island. We therefore designed the search to return a map of this two-phase landscape: a GOFEE global optimisation (Sec. 2.2) seeded from a flat reference layer, so that the flat basin is populated from the outset and any lower-lying basin the search finds is populated alongside it. The result is a sampled distribution over ΔZ and energy—an *exploration* density, not a thermodynamic density of states. Because the sampling is deliberately biased, we use it only for unweighted structural and energetic comparisons; the number of structures in a basin is a descriptive attribute of the sampled database, not a physical population.

Across the two models the search returns 1723 structures spanning ΔZ from 0.001 to 5.57 Å, with both phases populated in each (Table 3). Fig. 2 draws that landscape: every structure is placed at its own vertical span and at its energy above the model’s lowest, so the figure is the map the search was designed to produce rather than a selection of representative structures. The points form a continuous band that descends from the flat window into an island minimum near $\Delta Z = 3.7$ Å in both models, and the two panels share one energy scale, so a given height means the same relative energy in either.

The island is the ground state in both models, despite the flat bias

The lowest-energy structure found is an island in each model, with its global minimum at $\Delta Z = 3.65$ Å (Fe/MgO) and 3.77 Å (Fe-B/MgO) (Table 3, column 5)—a three-dimensional cluster rather than a flat film.

This result is *against* the direction of the bias: every search was seeded from a flat reference layer and, in the early phases, perturbed only at small scale from it, so the flat configuration had a systematic head start. The search nevertheless left the flat basin and converged on an island in both models. The flat-wetting state is therefore not an artefact of how the initial structures were built—it is genuinely higher in energy than the island. (MT-1)

The flat configuration is a distinct, higher-energy basin

Taking for each model the lowest-energy structure with $\Delta Z \leq 1.0$ Å isolates the flat basin’s best member. In both models it lies above the global minimum—by 0.1888 eV/atom for Fe/MgO and 0.1493 eV/atom for Fe-B/MgO (Table 3, column 6). The flat film is thus a separate basin of the landscape, not merely the high- ΔZ tail of a single minimum. (MT-2)

The branches are families, not single structures

The ΔZ values are continuous rather than grouped into discrete island heights: in each model the island branch spans $\Delta Z \approx 1\text{--}5.6$ Å with a median of 3.02 Å (Fe/MgO) and 2.58 Å (Fe-B/MgO), and the flat branch spans the whole window below 1 Å. The low-energy members of each branch are likewise not a single repeated structure: comparing the lowest-energy structures of each branch by their radial/angular fingerprint shows that each branch’s low-energy set splits into two distinct structural motifs, and that a motif recurs across independent searches—for Fe/MgO, the five lowest flat structures come from five different searches and fall into two motifs, four of them in the dominant one. (MT-8) The fingerprint is computed on the whole template-plus-film structure, so the fixed MgO template compresses the pair distances and the motif count is a lower bound on the structural diversity present; a film-resolved descriptor would be sharper. Pairwise distances within the low-energy set of a branch are 0.41–0.44 of that branch’s random-pair distance scale, i.e. the low-energy structures are consistently more compact than the branch as a whole. The count is also limited by how many searches contribute to the low-energy set: the Fe-B/MgO flat set within 0.02 eV/atom of its minimum comes from two searches and its island set from three, so those counts rest on fewer independent searches than the corresponding Fe/MgO numbers (five and four).

Experimental correspondence

The films in this study are about one monolayer thick on average—25 metal atoms per 25 substrate sites in the 5×5 cell, the boron-containing film carrying three additional boron atoms—and that is the regime in which the growth mode of Fe on MgO(001) has been characterised experimentally. At room temperature the observed growth is three-dimensional: He-atom scattering shows 3D metal island growth of Fe on MgO(001), with quantified island densities, sizes and shapes, suppressed only when the film is deposited at low temperature (140 K), where a monolayer almost completely covers the substrate [10]; scanning tunnelling microscopy finds that sub-nanometre Fe grows three-dimensionally on MgO, with island coalescence between 3.5 and 6.5 monolayers and a two-dimensional growth mode reported only above ≈ 6.5 monolayers [11]; and grazing-incidence small-angle X-ray scattering on five monolayers evaporated at room temperature likewise identifies Volmer–Weber growth with spherical islands [12]. A flat, continuous film is obtained instead by low-temperature deposition [10] or by slow deposition onto a cleaved, oxygen-annealed crystal, where the first monolayer grows pseudomorphically and layer by layer [13]. Our two basins therefore map onto the two outcomes that are experimentally accessible at this coverage—the flat basin onto the pseudomorphic monolayer and the island basin onto the 3D clusters—and the energy ordering we find places the flat, wetting configuration above the island. This is a consistency check rather than a prediction: the observed growth mode is also set by kinetics, so the correspondence is between structural configurations, not between energies. Because the growth mode turns two-dimensional only above ≈ 6.5 monolayers [11], the comparison is specific to the monolayer regime and should not be extrapolated to thick films.

3.2 The flat (wetting) phase with and without boron

With the two phases established, we compare how adding boron to the Fe film shifts the flat basin. Table 4 gives the flat-basin minimum dE/N for the two models, i.e. the same host with and without the metalloid,

Table 4: Flat-basin minimum dE/N (eV/atom) for the same Fe host with and without boron.

Film	Flat-basin min dE/N (eV/atom)	B effect
Fe	0.1888	—
Fe-B	0.1493	−0.040

and Fig. 3 places the two values side by side with the change between them marked.

Boron lowers the flat-state energy

Adding boron lowers the flat basin’s best member by 0.040 eV/atom—from 0.1888 to 0.1493 eV/atom—a 21 % reduction of the flat–island separation. (MT-3) The effect is measured across 13 independent Fe/MgO searches and 6 independent Fe-B/MgO searches; its search-level significance is given in the uncertainty paragraph below.

Boron does not bond to the MgO interface

Restricting to the low-energy window ($dE/N \leq 0.05$ eV/atom), only 1 of the 72 Fe-B/MgO structures in that window has any boron atom within 2.6 Å of an oxygen of the substrate; the maximum boron–oxygen contact fraction in the window is 0.33. Outside the window the picture is different—101 of the 471 remaining Fe-B/MgO structures do have a boron–oxygen contact—which is why the statement is restricted to the low-energy window rather than made over all sampled structures. Boron therefore remains inside the metal film rather than wetting the oxide, so its effect on the flat phase is film-internal rather than a change in interfacial bonding. (MT-7)

Uncertainty from the choice of search

Each model was searched from several independent random seeds (13 completed searches for Fe/MgO, 6 for Fe-B/MgO), and structures from one search are correlated with each other, so the independent unit is the search, not the structure. Taking each search’s own flat-basin minimum, the per-search minimum falls from 0.2171 eV/atom without boron to 0.1882 eV/atom with it on average (medians 0.2185 and 0.1834), a median shift of 0.0351 eV/atom with 94 % of resampled replicates agreeing in sign—the effect is not carried by a single favourable search, and the best-structure difference of Table 4 (−0.040 eV/atom) is the extreme of the same distribution rather than a lone outlier. A two-sided permutation test over the 19 searches gives $p = 0.0444$, significant at the 5 % level. In words, p is the probability that a random reshuffling of the 19 per-search minima into two groups of 13 and 6 would, by chance alone, produce a shift in the flat-state energy at least as large as the one observed—so $p = 0.0444$ means such a shift arises by chance in only about 4.4 % of reshufflings, i.e. the observed lowering of the flat state is unlikely to be a chance outcome of which searches happened to contain boron. The test is an exact enumeration of all 27 132 partitions of the pooled search minima, so this value is exact for the data and its resolution floor is $p = 3.7 \times 10^{-5}$ —more than three orders of magnitude below the value reported, i.e. the conclusion is not an artefact of too few searches.

3.3 The island (dewetting) phase and the separation between phases

Adding boron leaves the *island* phase qualitatively similar. In both models the lowest-energy structure is an island—at $\Delta Z = 3.65$ Å (Fe/MgO) and 3.77 Å (Fe-B/MgO) (Table 3)—and the island branch spans the same broad range of heights in both ($\Delta Z \approx 1$ –5.6 Å, median 3.02 Å without boron and 2.58 Å with it). Boron does not change what the dewetted state looks like; it changes how far above it the flat, wetting state sits.

Because each model is referenced to its own lowest energy, the quantity that compares the two phases is the separation between them—the flat-basin minimum of Table 4—and that separation is what boron reduces. The present data do not resolve *how* boron reduces it: a lower flat-basin energy could equally arise from stabilising the flat phase or from destabilising the island, and distinguishing the two would require the island phase to be referenced to an absolute (rather than per-model) energy scale, or a converged treatment

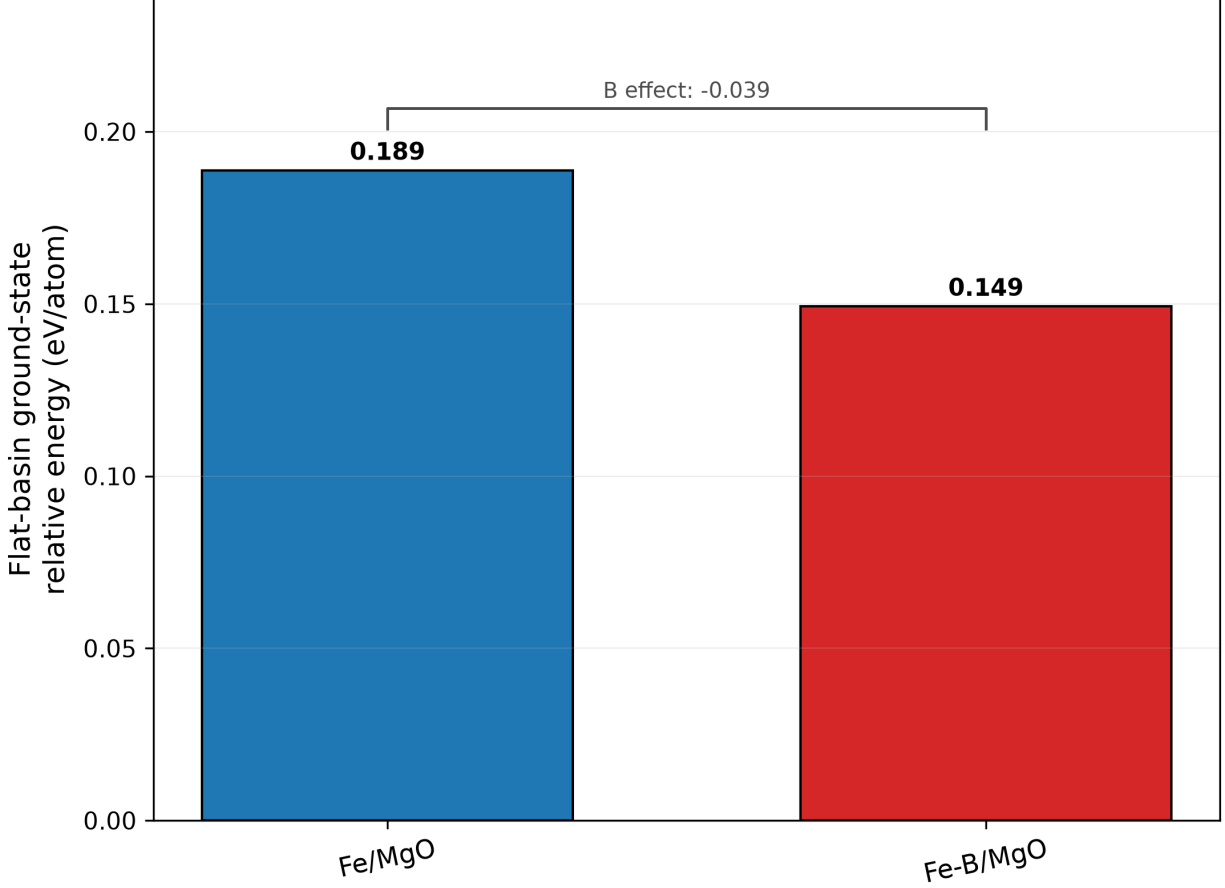


Figure 3: Flat-basin minimum relative energy for the same Fe host with and without boron. A bar is the lowest energy found among the structures of the flat window ($\Delta Z \leq 1.0$ Å), in eV/atom above the model’s own lowest energy, and the bracket gives the shift produced by boron.

of both basins. We therefore state the effect as a change in the flat–island separation, not as the stabilisation of either phase individually.

3.4 Summary of the results

The biased GOFEE search, seeded from a flat reference, maps both the flat (wetting) and island (dewetting) phases in both models, over $\Delta Z = 0.001$ – 5.57 Å. The island is the ground state in both models ($\Delta Z = 3.65$ and 3.77 Å), even though the search was biased toward the flat configuration—the flat bias did not manufacture the result—and the flat configuration is a distinct, higher-energy basin, 0.1888 eV/atom above the island for Fe/MgO and 0.1493 eV/atom for Fe-B/MgO, in both cases a separate basin rather than the high- ΔZ tail of a single minimum. Boron lowers the flat–island separation by 0.040 eV/atom (13 vs 6 completed searches, exact two-sided permutation $p = 0.0444$) without displacing the island as the ground state, and it does not bond to the MgO interface—it stays within the film, so its effect on the flat phase is film-internal.

The low-energy structures of each branch form a small set of recurring motifs rather than one repeated structure, and ΔZ is continuous: the two phases are families, and the energetics above are compared unweighted, since the sampled densities are biased and not physical (MT-8).

The mechanistic origin of these trends (electronic structure, interface registry) and the robustness of the method to its parameters are presented in the Supplementary Material.

4 Discussion

4.1 Why the island is the ground state

The search finds an island ground state in both models, with the flat film sitting 0.19 eV/atom higher for Fe/MgO and 0.15 eV/atom higher for Fe-B/MgO. The electronic-structure analysis (Supplementary Material) indicates why. In the flat monolayer every metal atom is registry-locked directly atop an oxygen of the MgO surface—the registry of the reference construction, and the one determined experimentally for the first monolayer of Fe on MgO(001) by LEED I–V analysis [13] and used in first-principles models of the Fe|MgO|Fe interface [3]—maximising Fe–O orbital overlap and pushing the Fe *d*-band centre down to –0.23 eV. The island abandons that registry: the number of metal atoms registered directly atop an oxygen drops from 25 (of 25) to 9 (of 25), the island’s interface Fe *d*-band centre rises to +0.51 eV (+0.60 eV averaged over all island Fe), and the film becomes electronically “quieter” (lower DOS at the Fermi level, weaker spin polarisation). The island’s energy gain is therefore dominated by reduced forced interfacial coupling and restored metal cohesion: the registry-locked monolayer spends its bonding on Fe–O contacts while forgoing the three-dimensional Fe–Fe coordination available to a cluster, and the island reverses that trade.

It is not a lattice-strain effect. The in-plane mismatch of this interface (3.6 % of the MgO lattice against the Fe lattice, Sec. 2.1) is carried by the substrate, which is built in that compressed state and held fixed, not by the film; the Fe film sits at its own equilibrium lattice constant and is not strained in-plane, so there is no film strain for the island to relieve. The mechanism is consistent with the weak Fe–MgO coupling found in first-principles treatments of the interface [3], and with the experimental observation that a flat monolayer requires low-temperature or slow deposition while room-temperature growth gives 3D islands at the same one-monolayer coverage [10–12].

4.2 Why boron stabilises the flat film

Boron lowers the flat-state energy by 0.040 eV/atom, bringing the flat configuration closer to the ground state. Two observations frame the mechanism. First, boron does not bond to the MgO interface—it remains inside the metal film (a single B–O contact in one of 72 windowed Fe-B structures)—so its effect is not interfacial. Second, the shift is not carried by one favourable search: the Fe-B model’s six per-search flat minima lie between 0.1493 and 0.2481 eV/atom, five of them below the Fe host’s median of 0.2185 eV/atom, and 94 % of resampled replicates agree in sign (Sec. 3.2). Both observations point to an intrinsic, film-internal role for boron rather than an interfacial or statistical one.

Confusion principle and amorphous formation

The flat film can be read as the disordered, amorphous-like configuration and the island as the ordered, crystalline-like one. Under this reading, the stabilisation of the flat state by added elements follows the confusion principle of metallic-glass formation [14]: the more elements in an alloy, the harder it is for the alloy to select a viable crystal structure, and the greater the tendency toward glass (amorphous) formation.

Our results are partly consistent with this. Adding boron lowers the flat-basin energy (0.1888 → 0.1493 eV/atom), i.e. the added element stabilises the disordered configuration relative to the ordered one, in the direction the principle predicts. The present models test this for a single added metalloid and cannot decide between the competing readings of the principle—whether it acts through the *presence of an added species* by a film-internal mechanism, or through the *number* of species, which would require the same host with two different additions to separate. We therefore read our result as consistent with the confusion principle acting through the added metalloid and leave the element-count reading untested here (see Sec. 4.4).

We note, however, that the boron-containing film retains the island as its ground state—the confusion principle stabilises the flat state but does not, in this small model system, fully suppress the ordered configuration. The precise origin of boron’s effect—whether it lowers the flat basin’s energy or raises the island’s—is not resolved by the present data and is a natural target for the re-relaxation and further analysis.

4.3 Implications for MTJ stacks

The performance of MgO-based magnetic tunnel junctions rests on the structural quality of the metal/oxide interface. The giant tunnel magnetoresistance of single-crystal Fe/MgO/Fe junctions arises from coherent spin-polarised tunnelling across a lattice-matched interface, with the residual mismatch accommodated by interfacial dislocations and the growth conditions chosen to minimise them [1]. The present results connect to this in two ways. First, they show that a perfectly flat, registry-locked metal layer maximises Fe–O bonding at the expense of metal coordination and is therefore energetically penalised relative to a clustered film—a consideration for interface engineering in MgO-based tunnel junctions, whose barrier must stay flat and coherently matched for the tunnelling magnetoresistance to reach its high values [1]. Second, they show that boron acts to stabilise the flat configuration without bonding to the interface, consistent with the picture of boron as a film-internal agent that promotes a flat, well-wetting interface. These are trend-level, model-system conclusions; quantitative transfer to a device stack would require the converged relaxations and a fuller treatment of the interface.

4.4 Limitations

One scope bound follows from the design rather than from the methods. The paper compares a single host with and without one added element, boron. Nothing here separates the effect of *boron* from the effect of *an added element as such*; that would need a second chemically different addition in the same host, which is not analysed (the element-count reading of the confusion principle is therefore untested—Sec. 4.2). What the design does exclude is a trivial explanation from the construction: the two models share the lattice constant, substrate, reference-layer geometry and candidate-generation schedule, and differ only by the boron (Sec. 2.6).

The results are qualitative/trend-level: the structures are not DFT-converged minima (residual forces $\sim 1\text{--}2$ eV/Å), the models are single-layer slabs at Γ -point sampling, and the flat/island split uses a chosen ΔZ threshold. The two models also have unequal seed counts (13 completed searches for Fe/MgO against 6 for Fe-B/MgO) and their atom counts differ (75 and 78), so sampling fractions are compared only as trends, not as like-for-like populations. The flat state is described as a higher-energy configuration, not a proven metastable state. A full re-relaxation of representative structures is necessary to place these conclusions on converged minima.

Two features of the model itself bound the interpretation. First, the strain convention is the inverse of the experimental stack: here the in-plane lattice is the Fe lattice constant and the MgO substrate is the compressed component, held fixed, whereas in a real junction the bulk MgO imposes its lattice on a thin Fe film, which absorbs the mismatch as in-plane strain and relieves it through interfacial dislocations [1]. The film in this work is therefore unstrained, and this model does not represent the strained-film situation. Second, both phases are built on a body-centred-cubic Fe lattice, while the experimentally reported structure of ultrathin Fe on MgO(001) is body-centred tetragonal below about 10 Å [13]; the island branch ($\Delta Z \approx 1\text{--}5.6$ Å) lies in that regime (Sec. 2.6). The flat–island comparison is therefore a trend obtained within one fixed lattice model, and extending it to thicker, experimentally strained films would require a different construction.

The search-level statistic rests on 19 searches, and it is asymmetric

The comparison pools 13 completed searches of the Fe/MgO model against 6 of the Fe-B/MgO model. The permutation test enumerates all 27 132 partitions of that pool, so its resolution floor is $p = 3.7 \times 10^{-5}$ and the reported $p = 0.0444$ is not limited by the number of searches; it would, however, tighten with more completed Fe-B/MgO searches, since the smaller group sets the width of the null. Two further bounds follow from the same asymmetry. The boron-free interface statement (Sec. 3.2) is measured on the 72 Fe-B/MgO structures inside the low-energy window, which come from four of the six searches—so its effective number of independent observations is closer to four than to 72—and the Fe-B/MgO motif counts (Sec. 3.1) rest on two (flat) and three (island) searches against five and four for Fe/MgO. Statements resting on those sets are correspondingly weaker evidence than the flat-basin energies of Table 4, which use every search of both models.

5 Conclusion

We studied the wetting of a metal film on MgO(001)—the interface at the heart of MgO-based magnetic tunnel junctions—for the same Fe host with and without added boron. Using a biased, surrogate-driven search that targets the two configurations the wetting question is about, we mapped the landscape into its two growth modes: a flat, well-wetting film and a dewetted island. In both models the island is the ground state, so the flat film is a distinct, higher-energy basin—by 0.1888 eV/atom for Fe/MgO and by 0.1493 eV/atom for Fe-B/MgO. Adding boron lowers the flat–island separation by 0.040 eV/atom, a 21 % reduction, at a search-level significance of $p = 0.0444$ —the fraction of random relabellings of the 19 searches that would reproduce a shift this large if boron had no effect. Boron acts inside the film rather than at the interface, and it does not displace the island as the ground state: the ordered, three-dimensional configuration remains the more stable of the two in both models.

Boron insertion therefore lowers the relative energy of the flat, well-wetting configuration, moving it closer to the island ground state without displacing it. This is the direction the confusion principle of metallic-glass formation predicts—an added element stabilises the disordered, flat configuration relative to the ordered one [14]—and it matters for MgO-based tunnel junctions, whose performance depends on a flat, coherent metal film on the barrier. The metal film’s tendency to dewet into islands at monolayer coverage, and the control of interface flatness that modern junction fabrication demands, make the energetic balance between these two configurations a quantity worth knowing.

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